

Fly GACA — Continuous Improvement Audit

SECTION 06-operations-it

TYPE audit

STATUS **active**

OWNER Founder

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Date: 2026-06-09 · Scope: Technical Code & Architecture, UI/UX & Web Design, Written Content & Communication, Product & Business Strategy.

This report applies a five-pillar continuous-improvement framework to the repo. Each finding is tagged **[code]** (actionable in this repository) or **[ops]** (needs external action — legal, billing keys, sales, analytics accounts). Findings were verified against source; one originally-reported UI finding was retracted as false (the hero grid already stacks on mobile at `landing.css:613`).

A first batch of verified, low-risk **[code]** fixes was applied alongside this report — see [Applied in this pass](#).

Priority summary

#	Pillar	Finding	Tag	Severity	Status
1	Content	Chat error/rate-limit/not-ready replies were English-only — broke Arabic UX on any backend failure (<code>assets/js/chat.js</code>)	code	High	Fixed
2	Technical	Silent <code>catch</code> on the entitlement read demotes every paid pilot during a Firestore outage with no log (<code>functions/index.js:60</code>)	code	High	Fixed
3	UI/UX	No <code>:disabled</code> button styling — disabled CTAs render fully active (<code>assets/css/base.css</code>)	code	Med	Fixed
4	Technical	47 <code>2.*</code> sync-artifact duplicate files committed and unreferenced	code	Low	Fixed
5	Technical	<code>guards.js</code> duplicated across <code>functions/rag/</code> and <code>captadel/src/brain/</code> with no CI parity check — can silently diverge	code	Med	Open
6	Technical	Rate-limiter wholesale <code>hits.clear()</code> on overflow under-counts a window (<code>functions/rag/ratelimit.js:53-62</code>)	code	Med	Open
7	Technical	<code>TESTER_EMAILS</code> (<code>assets/js/store.js</code>) ↔ <code>functions/staff.js</code> synced by hand, no CI guard	code	Med	Open
8	Technical	Monolithic page modules: <code>document.js</code> (~790 lines), <code>landing.js</code> (~629), <code>tools-aerodromes.js</code> (~508)	code	Med	Open
9	UI/UX	Nav touch targets undersized (<code>.lang-toggle</code> / <code>.nav-search</code> < 44px)	code	Med	Open
10	UI/UX	No visible loading/empty-state feedback in chat, library search, document reader	code	Med	Open

#	Pillar	Finding	Tag	Severity	Status
11	UI/UX	Form errors lack <code>role="alert"</code> / <code>aria-invalid</code> (pricing, account)	code	Med	Open
12	Content	Dynamic library/search result metadata not bilingual; <code>tools/</code> Uls partially un-translated	code	Med	Open
13	Content	Pricing page has no "prices subject to change / VAT" caveat	code	Low	Open
14	Content	"Captain Adel" / "AI flight instructor" / "study assistant" naming inconsistent	code	Low	Open
15	Product	Analytics dark — Cloudflare token empty, error beacon local-only, no product analytics (<code>assets/js/analytics.js</code>)	ops	High	Open
16	Product	No customer feedback loop (waitlist capture only)	ops	Med	Open
17	Product	Stripe integration coded but not wired to the pricing CTA (<code>functions/stripe.js</code>)	ops	Med	Open
18	Product	Legal entity + PDPL DPIA are hard gates on monetization / public launch (<code>phase0.md</code> , <code>roadmap.md</code>)	ops	High	Open
19	Product	B2B schools motion + social pipeline documented but not executed (<code>office/</code>)	ops	Med	Open

A. Technical Code & Architecture

Strong already: clean three-unit separation (static PWA / gateway / Captain Adel subtree); `*-core.js` pure modules with unit tests; defense-in-depth injection guards and XSS escaping (`chat.js` `esc()` / `safeUrl()`); region as a single source of truth (`functions/region.js`); comprehensive CI gates (chrome, data, links, e2e, eval, rules).

Findings

- **[#2, fixed]** `functions/index.js` `identify()` swallowed the entitlement read with an empty `catch ( _ )`. A sustained Firestore outage would silently treat every paid pilot as free with no signal. Now logs `logger.warn('entitlement read failed', ...)`; fail-to-free behaviour unchanged.
- **[#4, fixed]** 47 `2.*` files (`package 2.json` , `CLAUDE 2.md` , `functions/content 2.js` , 20+ `tools-* 2.js` , duplicated `docs/` , `tests/`) were committed sync artifacts, referenced by nothing. Removed.
- **[#5]** `functions/rag/guards.js` and `captadel/src/brain/guards.js` are ~99% identical by design (gateway + service copies) but kept in sync by hand. **Recommend** a CI guard that diffs the shared logic (or a generated shared module) so injection detection can't diverge.
- **[#6]** `functions/rag/ratelimit.js:53-62` clears the entire `hits` map when it exceeds `MAX_KEYS` , under-counting one window for all keys. Acceptable at MVP scale; **recommend** an LRU/segmented eviction before high traffic.

- [#7] `TESTER_EMAILS` in `assets/js/store.js` must mirror `functions/staff.js`; a missed update mis-grants access. **Recommend** a small CI check asserting parity.
- [#8] Decompose the largest page modules (`document.js`, `landing.js`, `tools-aerodromes.js`) along their internal seams (reader vs bookmarks vs i18n dict).

B. UI/UX & Web Design

Strong already: WCAG-AA+ contrast tokens (`tokens.css`); fluid `clamp()` type scale; skip links, focus-visible rings, ARIA landmarks and live regions; logical-property RTL; responsive breakpoints at 1024/900/760/480.

Findings

- [#3, fixed] No `:disabled` button rule existed — disabled CTAs looked active. Added `.btn:disabled` (opacity .5, `not-allowed`, no transform/pointer-events) to `base.css`.
- [#9] `.lang-toggle` / `.nav-search` compute to ~28–32px tall — below the 44px touch target. **Recommend** bumping vertical padding (verify visually to avoid nav reflow).
- [#10] Add a spinner/skeleton + bilingual "loading..." to chat, library search and the document reader; users on slow links get no feedback today.
- [#11] Add `role="alert"` / `aria-invalid` to form validation so screen readers announce failures (pricing, account/waitlist forms).
- *Retracted:* "hero grid doesn't stack on mobile" — false; `landing.css:613` stacks it at `max-width:760px`.

C. Written Content & Communication

Strong already: identical not-affiliated / verify-against-GACA disclaimer strip on every page + full `about.html`; BLUF marketing copy; calm instructor-grade tone with no hype; the Captain Adel persona handles Arabic/English code-switching.

Findings

- [#1, fixed] `chat.js` `NOT_READY` / `ERROR` / `RATE_LIMITED` were English-only. They now carry `{ en, ar }` and a `msg()` helper that reads `document.documentElement.lang` (set by the `landing.js` bilingual engine), so an Arabic-preferring pilot gets Arabic fallbacks when the engine is unreachable.
- [#12] Dynamic text injected by `library.js` / `doc-search.js` (result counts, filter state, "no results") and several `tools/` UIs are not bilingual. **Recommend** routing all injected strings through the same `data-en` / `data-ar` (or a small lang helper) so `check:i18n` semantics extend to runtime content.
- [#13] Add a small "prices for reference, subject to change; VAT per local law" caveat to `pricing.html`.
- [#14] Standardize the assistant's first-mention framing, e.g. "Captain Adel — an AI flight instructor (not a real person), trained on the GACAR."

- **Process:** ROADMAP Phase 4 flags a pending native-Arabic review; current AR copy (including the strings added in this pass) should pass through it.

D. Product & Business Strategy

Strong already: painkiller fit (regulatory fragmentation), defensible moats (GACAR corpus depth, cited-to-Part AI, native bilingual), coherent freemium + Pro + Exam-term + school-seat + pack pricing, transaction-safe cross-platform entitlements that never clobber. These are **[ops]** items — recorded here, not changed in code.

Findings

- **[#15]** Instrument before launch: the Cloudflare Web Analytics token in `assets/js/analytics.js` is empty and the error beacon is local-only. Stand up a privacy-friendly product analytics tool now to capture a pre-paywall baseline.
- **[#16]** Add a feedback loop (in-app "report an issue" → ticket) before paying customers arrive.
- **[#17]** Wire the `pricing.html` CTA to the already-coded `functions/stripe.js` checkout (sandbox-testable without the legal entity).
- **[#18]** Legal entity registration (`phase0.md` P0-3) gates monetization; the PDPL DPIA gates the public Phase 3 launch. Start both in parallel with engineering.
- **[#19]** Execute the documented B2B schools outreach (`office/gtm-schools.md`) and publish the ready social launch posts (`office/social-media/`).

Applied in this pass

Low-risk, verified **[code]** fixes shipped with this report:

1. **Bilingual chat fallbacks** — `assets/js/chat.js` (`MESSAGES` + `msg()`).
2. **Entitlement-read observability** — `functions/index.js` (`logger.warn` , behaviour unchanged).
3. **Disabled-button styling** — `assets/css/base.css` (`.btn:disabled`).
4. **Repo hygiene** — removed 47 unreferenced `2.*` sync-artifact files.

Everything else above is left as a tracked recommendation, deliberately not implemented in this pass because it is larger, needs visual review, or is an ops/strategy action.